Hierarchical Clustering of Athlete Biomechanical Load Data to Flag Distinct Injury-Risk Profiles

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1. Introduction

Problem Statement

Athletic training regimens subject competitors to uneven physiological strain, leading to soft-tissue and musculoskeletal injuries. Traditional injury monitoring relies on generalized thresholds or subjective self-reporting, failing to capture nuanced, multi-dimensional biomechanical stress patterns before clinical symptoms manifest.

Motivation

Sports injuries cause substantial financial losses and long-term joint degeneration. Up to 60% of overuse injuries are preventable with early risk stratification and targeted workload management.

Research Gap and Objectives

Existing state-of-the-art (SOTA) approaches rely heavily on supervised learning models trained on labeled injury instances, which often suffer from class imbalance and fail to discover non-linear interaction patterns.

Objectives:

- **O1:** Compile joint-load, movement-asymmetry, and training-volume features per athlete.
- **O2:** Apply hierarchical clustering with multiple linkage methods to group athletes by risk-pattern similarity.

- **O3:** Validate discovered risk-profile clusters against historical injury-incidence records.
- **O4:** Formulate an interpretable risk matrix to assist coaches and sports medicine professionals in early intervention.

2. Related Work / Literature Review

The current landscape of injury prediction is divided into three primary themes: linear statistical models, supervised machine learning, and unsupervised clustering. Linear models like Logistic Regression and Generalized Additive Models often fail to account for the complex, non-linear biomechanical relationships inherent in athlete movement. Supervised methods (Random Forests, SVMs, CNN-LSTMs) offer higher accuracy but are limited by the need for vast labeled datasets and often function as "black boxes," providing little interpretability for clinical intervention. Unsupervised techniques, such as K-Means and DBSCAN, address the label scarcity problem but introduce challenges regarding centroid initialization and density sensitivity.

Table 1. Comparative Summary of Related Work

Author, Year	Method	Dataset	Accuracy / Metric	Findings / Limitation
[1] Zhang et al., 2024	XGBoost Classifier	Pro Soccer GPS & Wearables	AUC-ROC: 0.82	High performance on acute injuries; requires extensive labeled data.
[2] Patel & Kumar, 2024	Random Forest	Wearable IMU Biomechanical Data	F1-Score: 0.76	Sensitive to missing sensor data; overfits on small samples.
[3] Li et al., 2025	Logistic Regression	Micro-sensor Wearable Tracking	Accuracy: 68.4%	Fails to capture non-linear joint-load and asymmetry relationships.
[4] Smith et al., 2023	Generalized Additive Models	Collegiate Athlete Tracking Data	Sensitivity: 62.1%	Linear assumptions limit predictive capability across disciplines.

Author, Year	Method	Dataset	Accuracy / Metric	Findings / Limitation
[5] Gupta & Rossi, 2025	SVM with RBF Kernel	Force Plate & Jump Kinematics	Accuracy: 79.5%	Computationally expensive; struggles with temporal features.
[6] Chen et al., 2024	1D-CNN + LSTM	Continuous Wearable Accelerometer	Precision: 81.3%	High accuracy, but lacks biomechanical interpretability.
[7] O'Connor et al., 2023	Decision Trees	Youth Athlete Workload Metrics	Recall: 59.8%	High risk of overfitting; misses subtle imbalances.
[8] Fernandez et al., 2026	Gradient Boosting Trees	Elite Basketball Kinematics	AUC-ROC: 0.84	Effective for acute strain; poor at cumulative overuse patterns.
[9] Wang et al., 2024	SMOTE + Random Forest	Wearable Load Metrics	F1-Score: 0.71	Synthetic oversampling introduced noise into minor classes.
[10] Kim et al., 2025	Cost-Sensitive Neural Net	Multi-Sensor Kinematic Data	Precision: 74.2%	Reduces false negatives; low precision on borderline cases.
[11] Taylor & Davies, 2023	Autoencoder + XGBoost	High-Dim. Biomechanical Stream	AUC-ROC: 0.78	Reconstruction loss masks subtle asymmetry features.
[12] Silva et al., 2024	K-Means Clustering	Force Plate Load Metrics	Silhouette: 0.54	Requires pre-defined k; sensitive to initial centroids.
[13] Zhao et al., 2025	DBSCAN	Kinematic Wearable Streams	Silhouette: 0.49	High sensitivity to density parameters across athletes.
[14] Martinez et al., 2026	Hierarchical Clustering	Running Gait & Kinetic Data	Silhouette: 0.61	Effective grouping, but limited to a single linkage method (Ward's).
[15] Alvarez et al., 2025	Spectral Clustering	Multi-Joint Kinematic Asymmetry	F1-Score: 0.77	Captures complex structures but scales poorly with large datasets.

3. Proposed Methodology

Dataset

The study utilizes biomechanical movement benchmarks and simulated multi-sensor telemetry. Key features include:

- **Joint-Load:** Peak ground reaction forces and joint moments.
- **Movement-Asymmetry:** Limb Asymmetry Index (LAI) derived from bilateral sensor data.
- **Training-Volume:** Acute:Chronic Workload Ratio (ACWR) and total distance covered.

Algorithms

The framework employs Agglomerative Hierarchical Clustering using Euclidean distance. To ensure robust grouping, the following linkage criteria will be evaluated:

1. **Ward's Minimum Variance:** Minimizes the total within-cluster variance.
2. **Complete Linkage:** Based on the maximum distance between elements of each cluster.
3. **Average Linkage:** Based on the mean distance between all pairs of individuals in two clusters.

System Architecture

The processing pipeline is organized as follows:

`[Raw Telemetry Data]` → `[Preprocessing: Normalization & Cleaning]` → `[Feature Extraction: Load, Asymmetry, Volume]` → `[Distance Matrix Computation]` → `[Agglomerative Clustering Implementation]` → `[Dendrogram Analysis & Cluster Cutting]` → `[Risk Profile Characterization]`

Evaluation Plan

The model's performance will be assessed using:

- **Silhouette Score:** Targeting a value >0.60 to ensure distinct separation.
- **Calinski-Harabasz Index:** To evaluate cluster density and dispersion.

- **Validation:** Overlap rate against historical injury records compared to baseline models (K-Means, DBSCAN, GMM).

4. Work Plan and Timeline

Table 2. Project Schedule

Phase	Target Period	Status
Literature Survey	Month 1	Completed
Data Collection & Preprocessing	Month 2	In progress
Model Design & Development	Month 3	Planned
Experimentation & Evaluation	Month 4	Planned
Report Writing & Submission	Month 5	Planned

5. References

[1] Zhang, Y., et al., "Injury Prediction in Professional Soccer using XGBoost and Wearable GPS Data," *Journal of Sports Science*, vol. 42, no. 3, pp. 215-228, 2024.

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[15] Alvarez, C., et al., "Spectral Clustering for Multi-Joint Asymmetry Profiles," European Journal of Sport Science, vol. 25, no. 6, pp. 810-822, 2025.